

Skills Summary

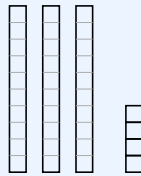
Subtracting Two-Digit Numbers with Place-Value Blocks

Use this reference while completing your worksheet or when studying.

Skill 1 — Subtracting without opening a rod

Rule: A tall rod is one ten; a small square is one.

Take the ones away first, then the tens. If the ones you have are enough, nothing needs opening — cross out and count what is left.



Example: Work out $58 - 24$ with blocks.

Step 1. Before touching anything I check the ones: there are more ones here than I need to take away, so no rod has to be opened and I can just cross out.

Step 2. Cross out four of the eight single squares, leaving four ones. Then cross out two of the five rods, leaving three rods. Three tens and four ones.

34

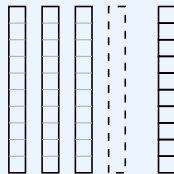
Try it: Work out $76 - 43$ with blocks.

check: 33

Skill 2 — Subtracting when you must open a rod

Rule: When there are not enough ones, *open* one rod into ten ones. The amount does not change — only how it is shown.

After opening: one rod fewer, ten ones more. Now take the ones away, then the tens.



Example: Work out $45 - 18$ with blocks.

Step 1. The ones I have are fewer than the ones I must take away, so the first move is not subtraction at all — it is opening a rod to get more ones.

Step 2. Opening one rod leaves three rods and fifteen ones. Take eight ones away, leaving seven; take one rod away, leaving two rods. Two tens and seven ones.

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Try it: Work out $73 - 26$ with blocks.

check: 47

Skill 3 — Estimating a difference, then subtracting

Rule: To estimate to the nearest ten, round *both* numbers first, then subtract the rounded ones.

Look at the ones digit: below five rounds down, five or more rounds up.

The estimate is a check. If your exact answer lands far from it, something has gone wrong in one of the two.

Example: Estimate $62 - 29$ to the nearest ten, then subtract exactly.

Step 1. An estimate is only worth doing if it is fast, so I round both numbers to tens and subtract those, keeping the exact work as a separate second pass.

Step 2. Rounding gives sixty and thirty, and taking one from the other gives the estimate. Exactly: two ones cannot give up nine, so open a rod, leaving five rods and twelve ones; take nine ones, leaving three, then take two rods, leaving three.

estimate **30** exact **33**

Try it: Estimate $85 - 38$ to the nearest ten, then subtract exactly.

check: estimate **50**, exact **47**